



北京邮电大学
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BUPT Compilers Lab4 2024Fall

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1 Code

1.1 lex.l

```
%{
    #include "syntax.tab.h"
}%
%option noyywrap

unic u[0-9a-fA-F]{4}
esc \\([\\\"\\/bfnrt]|{unic})
scp [^\\\"\\\x00-\x1f]
string \"({esc}|{scp})*\"

int 0|[1-9][0-9]*
frac \.[0-9]+
exp [Ee][+-]?[0-9]+
number -?{int}{frac}?{exp}?
leadingzero 0{int}

empty [ \\n\\r\\t]

%%

"{" { return LC; }
"}" { return RC; }
"[" { return LB; }
"]" { return RB; }
":" { return COLON; }
"," { return COMMA; }

"true" { return TRUE; }
"false" { return FALSE; }
"null" { return VNULL; }

{string} { return STRING; }
{number} { return NUMBER; }
{leadingzero} {return LEADINGZERO;}

{empty} {}

. { printf("lexical error: %s\\n", yytext); }
```

1.2 syntax.y

```
%{
```

```

#include"lex.yy.c"
void yyerror(const char*);
%}

%token LC RC LB RB COLON COMMA
%token STRING NUMBER LEADINGZERO
%token TRUE FALSE VNULL
%%

Json:
    Value
    | Json COMMA error { puts("Comma after the close, recovered");}
    | Json RB error { puts("Extra close, recovered");}
    ;
Value:
    Object
    | Array
    | STRING
    | NUMBER
    | TRUE
    | FALSE
    | VNULL
    ;
Object:
    LC RC
    | LC Members RC
    | Object Value error { puts("Extra value after close, recovered");}
    | LC Values RC error { puts("Comma instead of colon, recovered");}
    | LC Member COMMA error { puts("Comma instead if closing brace, recovered");}
    ;
Members:
    Member
    | Member COMMA Members
    | Members COMMA error { puts("Extra comma, recovered");}
    ;
Member:
    STRING COLON Value
    | STRING COLON COLON Value error { puts("Double colon, recovered");}
    | STRING COLON LEADINGZERO error { puts("Numbers cannot have leading zeroes, recovered");}
    | STRING Value error { puts("Missing colon, recovered");}
    ;
Array:
    LB RB

```

```

| LB Values RB
| LB Values RC error { puts("unmatched right bracket, recovered"); }
| LB Members RB error { puts("Colon instead of comma, recovered"); }
| LB Values error { puts("Unclosed array, recovered"); }
;
Values:
    Value
| Value COMMA error { puts("extra comma, recovered"); }
| Value COMMA COMMA error { puts("double extra comma, recovered"); }
| Value COMMA Values
| COMMA Values error { puts("missing value, recovered"); }
;
%%

void yyerror(const char *s){
    printf("syntax error: ");
}

int main(int argc, char **argv){
    if(argc != 2) {
        fprintf(stderr, "Usage: %s <file_path>\n", argv[0]);
        exit(-1);
    }
    else if(!(yyin = fopen(argv[1], "r"))) {
        perror(argv[1]);
        exit(-1);
    }
    yyparse();
    return 0;
}

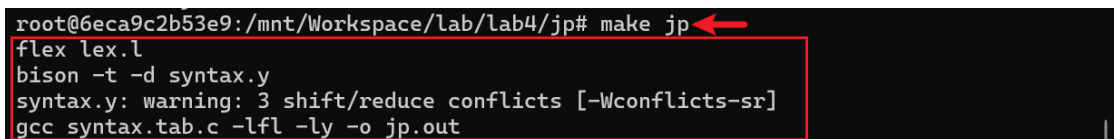
```

2 Run

The commands were used:

```
make jp
python3 jsonparser_test.py
```

The screenshot of the compilation and execution results.



```

root@6eca9c2b53e9:/mnt/Workspace/lab/lab4/jp# make jp
flex lex.l
bison -t -d syntax.y
syntax.y: warning: 3 shift/reduce conflicts [-Wconflicts-sr]
gcc syntax.tab.c -lfl -ly -o jp.out

```

```

root@6eca9c2b53e9:/mnt/Workspace/lab/lab4/jp# python3 jsonparser_test.py
For file fail82.json:
-----
["Unclosed array"]
-----
syntax error: Unclosed array, recovered
#####
For file fail04.json:
-----
["extra comma",]
-----
syntax error: extra comma, recovered
#####
For file fail05.json:
-----
["double extra comma",,,]
-----
syntax error: double extra comma, recovered
#####
For file fail06.json:
-----
[ , "c-- missing value"]
-----
syntax error: missing value, recovered
#####
For file fail07.json:
-----
["Comma after the close"],
-----
syntax error: Comma after the close, recovered
#####
For file fail08.json:
-----
["Extra close"]]
-----
syntax error: Extra close, recovered
#####
For file fail09.json:
-----
{"Extra comma": true,}
-----
syntax error: Comma instead if closing brace, recovered
#####
For file fail10.json:
-----
{"Extra value after close": true} "misplaced quoted value"
-----
syntax error: Extra value after close, recovered
#####
For file fail13.json:
-----
{"Numbers cannot have leading zeroes": 013}
-----
syntax error: Numbers cannot have leading zeroes, recovered
#####
For file fail19.json:
-----
{"Missing colon" null}
-----
syntax error: Missing colon, recovered
#####
For file fail20.json:
-----
{"Double colon": null}
-----
syntax error: Double colon, recovered
#####
For file fail21.json:
-----
{"Comma instead of colon", null}
-----
syntax error: Comma instead of colon, recovered
#####
For file fail22.json:
-----
{"Colon instead of comma": false}
-----
syntax error: Colon instead of comma, recovered
#####
For file fail32.json:
-----
{"Comma instead if closing brace": true,
-----
syntax error: Comma instead if closing brace, recovered
#####

```

```

For file fail33.json:
-----
["mismatch"}
-----
syntax error: unmatched right bracket, recovered
#####
Recovered/Total: 15/15
root@6eca9c2b53e9:/mnt/Workspace/lab/lab4/jp#

```